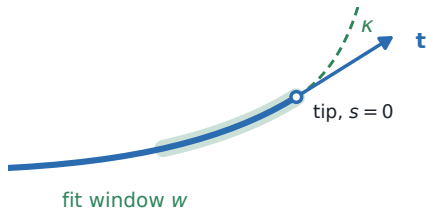
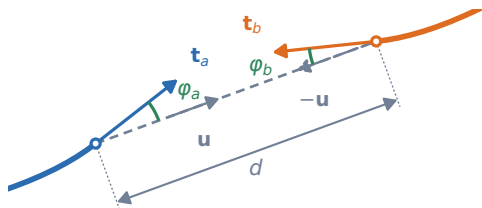


(a) Frame at a stub



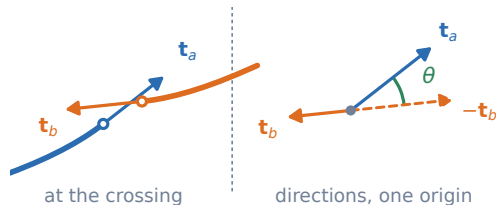
Each coordinate is fitted by least squares in arclength, tip at $s = 0$, then read outward.
Quadratic if the span is ≥ 18 px, else linear.
 $w = 24$ px locally, 55 px along a chain.

(c) Chord terms: $\varphi_a = 18^\circ$, $\varphi_b = 14^\circ$ here



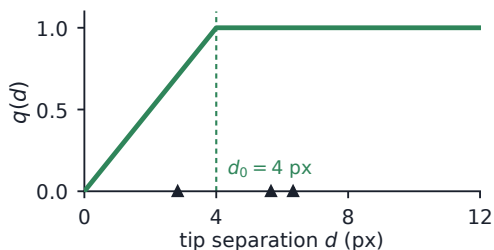
The two turns onto the chord u joining the tips. Unlike θ this sees lateral offset, so parallel arms that never meet score well on θ and badly here.

(b) Direct term: $\theta = 32^\circ$ here



$\theta = \arccos(\mathbf{t}_a \cdot -\mathbf{t}_b)$ compares two directions and never uses the chord, so it still means something when the tips are three pixels apart.

(d) Chord confidence $q(d)$



Inside a crossing the chord spans a few pixels and is badly quantised, so q fades the φ term out. Triangles: the six real tip separations at the crossing of Fig. 5.

$$C_{ab} = w_d \theta + w_t q(d) (\varphi_a + \varphi_b) + w_\ell d / \ell_0 + 30 w_\kappa |\kappa_a + \kappa_b|$$

junction: $w_d = 0.6$, $w_t = 1.2$, $w_\kappa = 1.0$, $d_0 = 4$ px, $w_\ell = 0$

gap: $w_d = 0.2$, $w_t = 1.0$, $w_\kappa = 0.5$, $d_0 = 3$ px, $w_\ell = 0.15$, $\ell_0 = 60$ px